SOP: Task 13 – Setup Monitoring Alerts and Action Groups

Objective

To configure Azure Monitor Alert Rules and Action Groups for Azure Virtual Machines in order to proactively monitor resource performance and automatically generate notifications when CPU utilization exceeds a predefined threshold.

Prerequisites

The following resources were already deployed and monitored before configuring alerts:

- Resource Group: **rg-mayank**
- Virtual Machine: **vm-1**
- Virtual Machine: **vm-2**
- Azure Monitor Enabled
- Azure Monitor Agent Installed
- Data Collection Rules (DCR) Configured
- Azure Monitor Workspace Configured
- Performance Metrics Collection Enabled
- Traffic Manager Profile
- Azure Subscription

Step 1: Verify Existing Monitoring Infrastructure

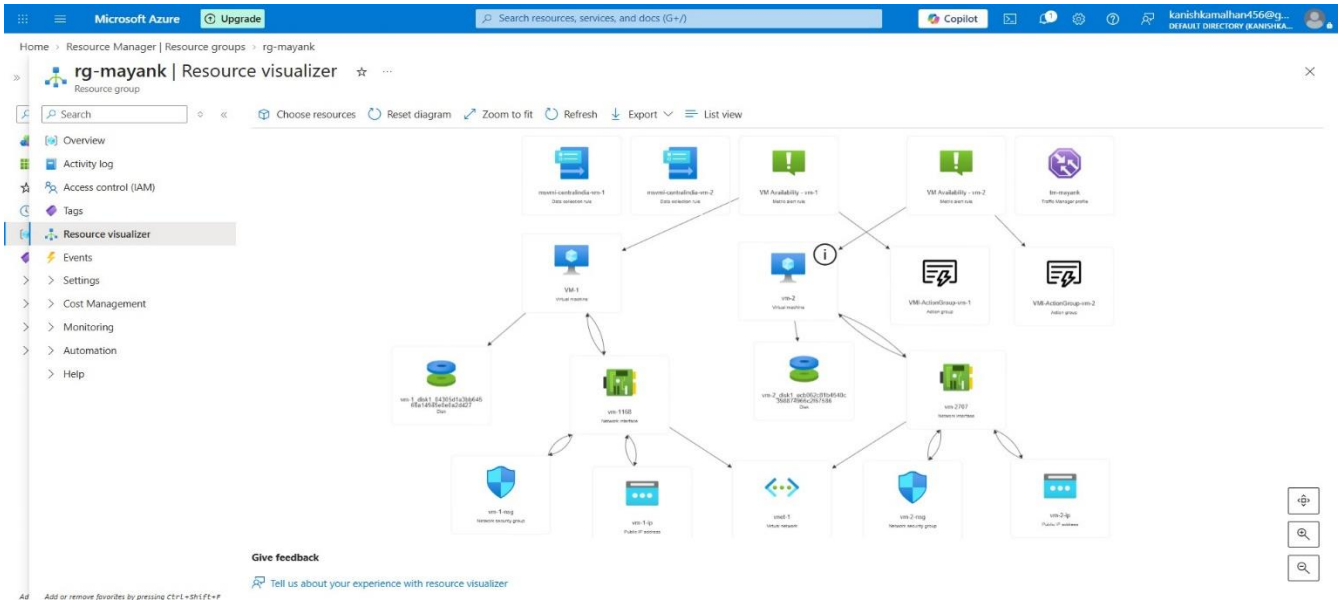
Navigate to: Azure Portal

→ Resource Group
→ rg-mayank
→ Resource Visualizer

Review the deployed resources and monitoring components available in the environment.

Observation

The infrastructure was already onboarded to Azure Monitor and contained the required monitoring resources necessary for configuring alert rules and notifications.



Step 2: Open Azure Monitor Alerts

Navigate to: Azure Portal

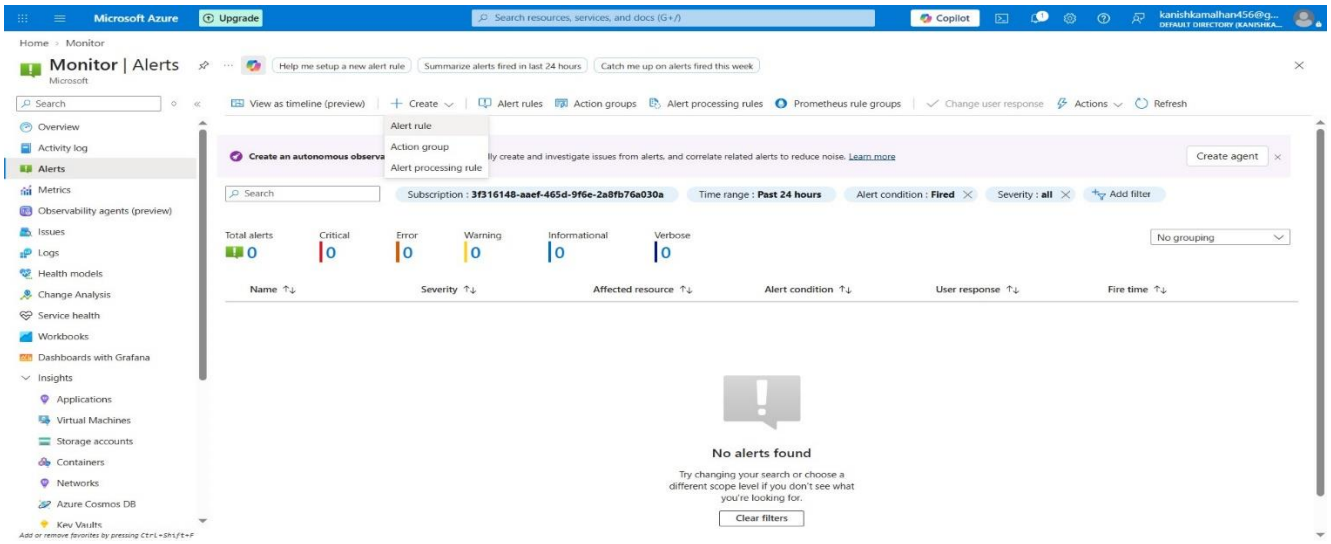
→ Monitor
→ Alerts

Click: Create

→ Alert Rule

Observation

The Azure Monitor Alert Rule creation wizard was successfully launched.



Step 3: Create an alert and Select Monitoring Scope

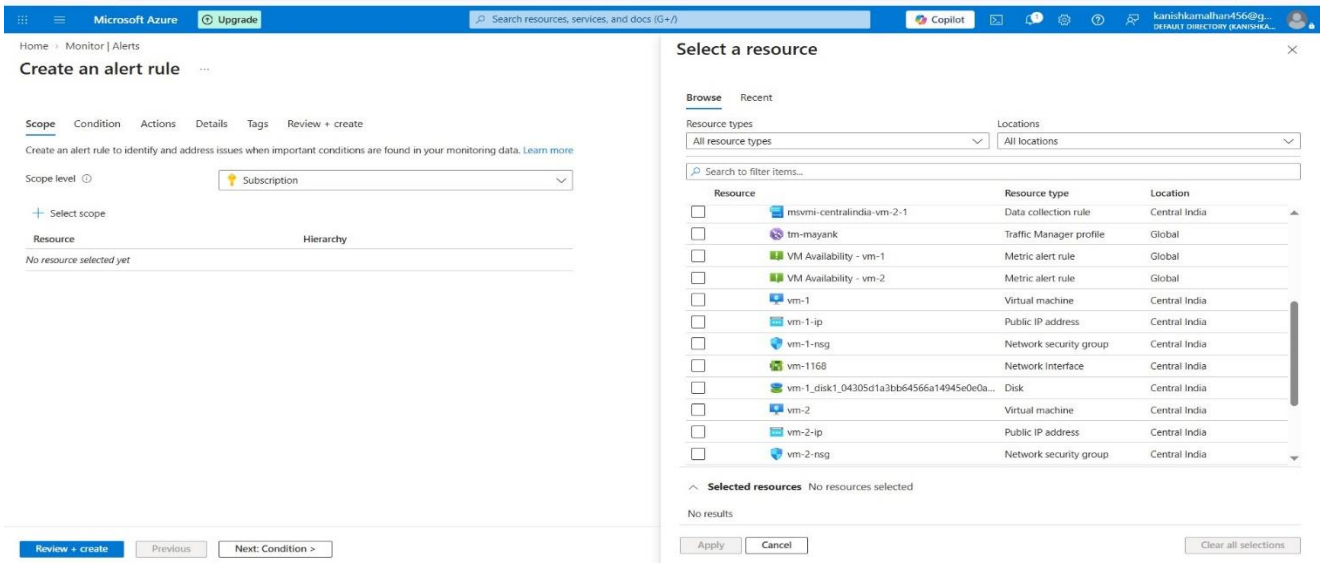
Under the Scope tab, click:

Select Scope : vm-1

Click: Apply

Observation

VM-1 was selected as the target resource for alert monitoring.



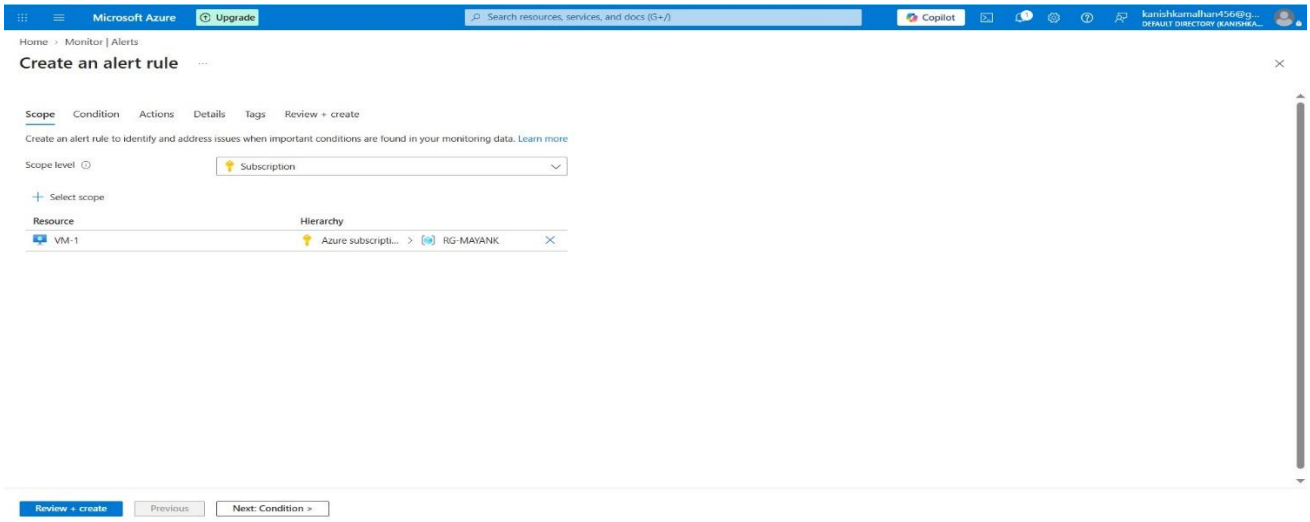
Step 4: Verify Scope Configuration

Review the selected resource details.

Resource : vm-1
Resource Group: rg-mayank
Subscription : Azure Subscription 1

Observation

The alert rule scope was successfully configured for VM-1.



Step 5: Configure Alert Condition

Navigate to: Condition

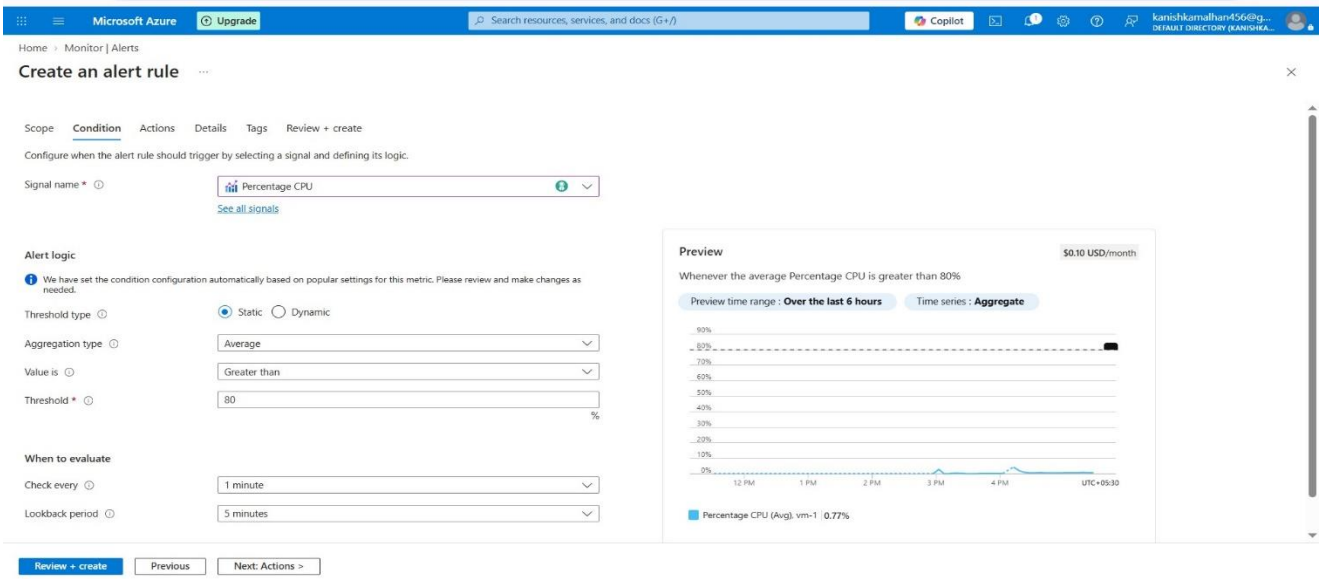
Select the monitoring signal:

Percentage CPU

Configure the alert logic:

Observation

The alert was configured to trigger whenever the average CPU utilization exceeds 80%.



Step 6: Configure Action Group

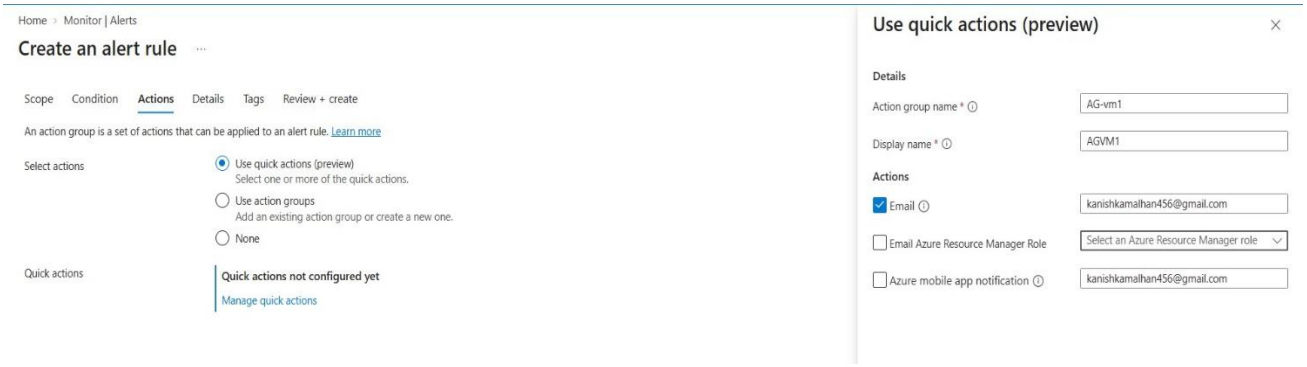
Navigate to: Actions

Select: Use Quick Actions

Provide the notification email address.

Observation

An Action Group was successfully configured to send notification emails whenever the alert condition is triggered.



Step 7: Configure Alert Rule Details

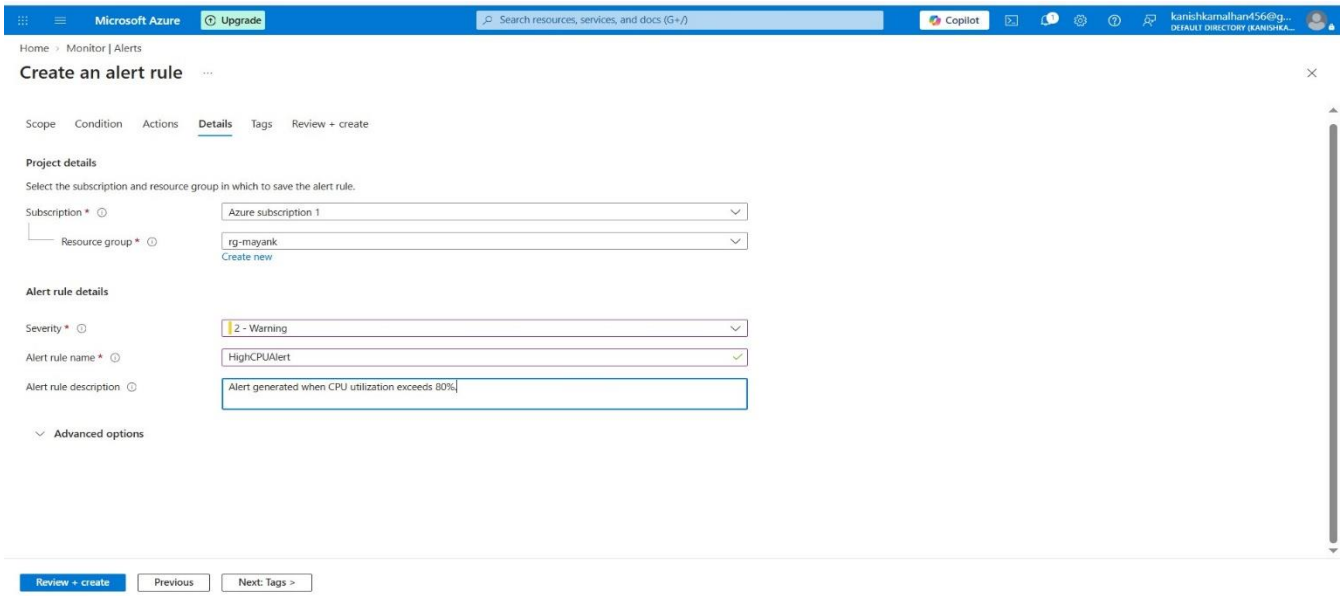
Navigate to: Details

Click: Review + Create

- Create

Observation

The alert rule configuration was completed successfully.



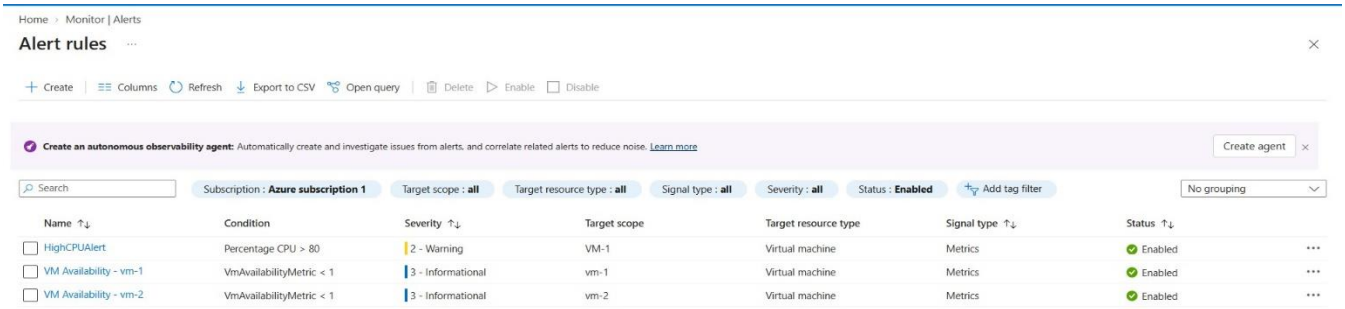
Step 8: Verify Alert Rule Creation

Navigate to: Monitor

→ Alerts
→ Alert Rules

Observation

The HighCPUAlert rule was successfully created and enabled for monitoring CPU utilization.



Step 9: Validate Monitoring Metrics

Navigate to: Monitor

→ Metrics

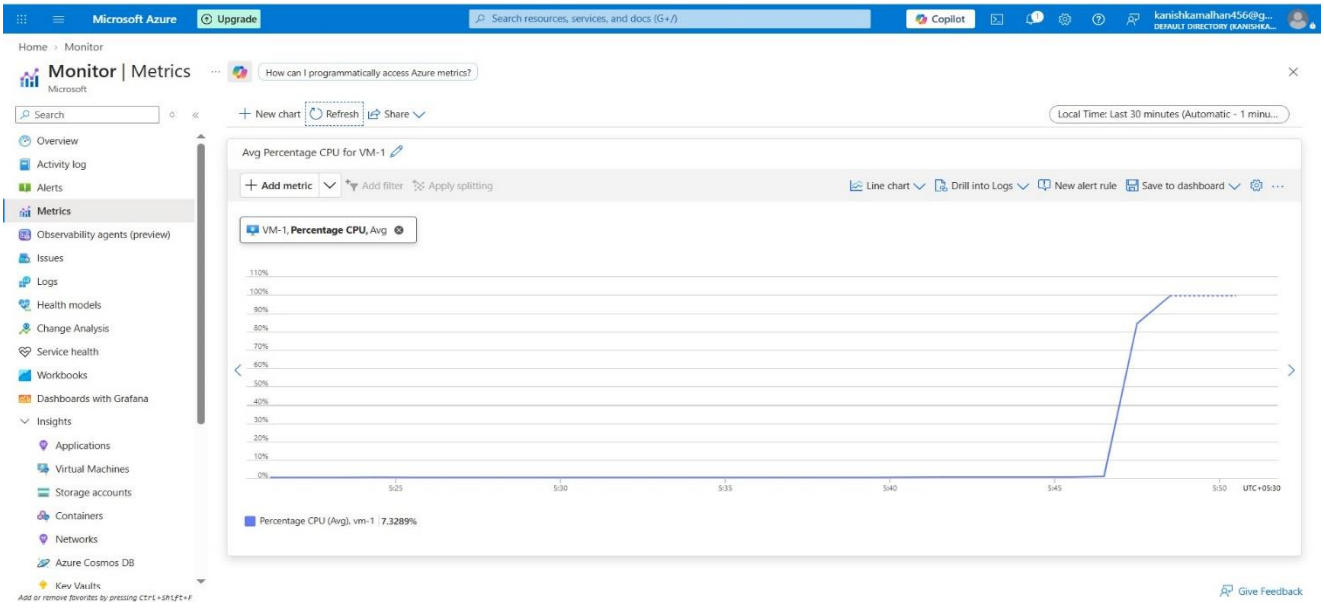
Select:

Resource : VM-1
Metric : Percentage CPU

Generate CPU load on VM-1 and monitor the graph.

Observation

CPU utilization increased significantly and exceeded the configured threshold value of 80%, reaching approximately 100%, confirming that the alert criteria were met.

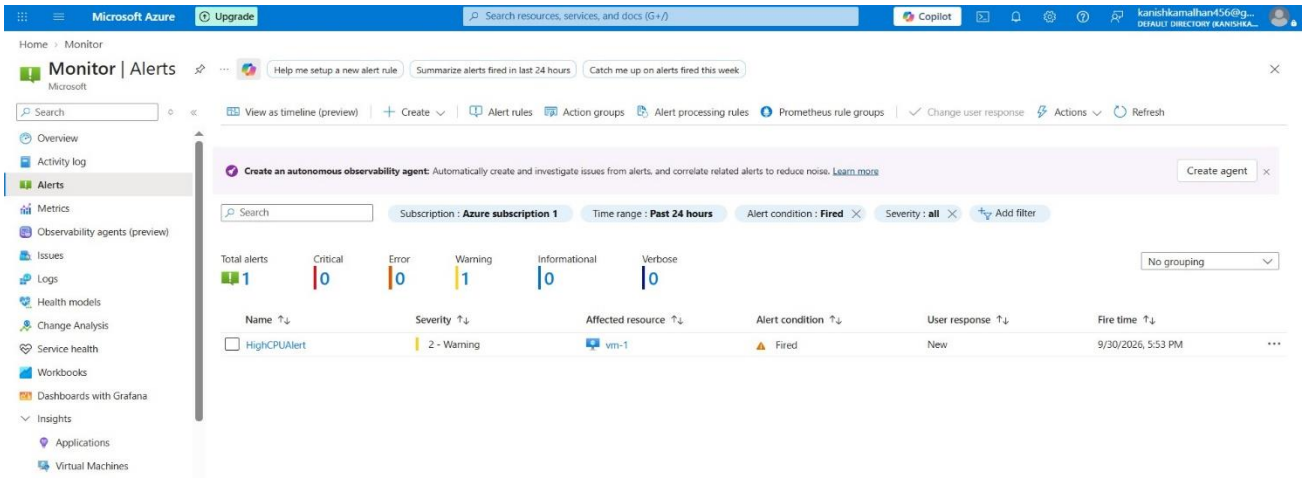


Step 10: Verify Alert Trigger

Navigate to: Monitor
→ Alerts
Review the fired alerts.

Observation

Azure Monitor successfully detected the CPU threshold breach and automatically triggered the HighCPUAlert alert rule.



Final Observation

Azure Monitor Alerts and Action Groups were successfully configured for VM-1. A metric-based alert rule was created to monitor CPU utilization and trigger whenever the average CPU usage exceeded 80%. An Action Group was configured to send email notifications. After generating high CPU utilization on VM-1, Azure Monitor detected the threshold breach and successfully fired the alert. Resource Visualizer confirmed the integration of monitoring resources, alert rules, action groups, and data collection rules within the monitored environment.

Result

Successfully implemented **Monitoring Alerts and Action Groups** using Azure Monitor. A CPU-based alert rule (**HighCPUAlert**) was configured for VM-1 and linked with an Action Group (**AG-vm1**) for automated notifications. Azure Monitor successfully monitored CPU utilization, detected threshold violations, and generated alerts, ensuring proactive monitoring and operational visibility for the Azure environment.